

Computational Physics: Materials Science

Exercise 4

Molecular dynamics of a sodium cluster

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Molecular Dynamics of a Sodium Cluster

This exercise simulates a 27-atom sodium cluster interacting through the Morse potential using classical molecular dynamics with the **Velocity-Verlet** integrator under periodic boundary conditions. Three thermostatting schemes (none, simple velocity rescaling, Berendsen) are compared, the melting transition is identified at 100 K, 300 K, 700 K and 1200 K and visualised in Ovito, and the simulation is repeated for copper. An AI assistant (Claude Sonnet 4.6) assisted with coding; all physical interpretation is independent.

Implementation

Goal, Units, and Potential

The pair potential is the Morse form (Girifalco & Weizer, Phys. Rev. 114, 687, 1959):

$$V(r) = D \left[e^{-2\alpha(r-r_0)} - 2e^{-\alpha(r-r_0)} \right]. \quad (1)$$

All calculations use **Å, fs, u, eV**. With $1 \text{ u} \cdot (\text{Å}/\text{fs})^2 = 103.6427 \text{ eV}$ the kinetic energy is $K = \frac{1}{2}m \sum v^2 \cdot 103.6427$ and temperature follows from equipartition, $T = 2K/(3Nk_B)$, $k_B = 8.617 \times 10^{-5} \text{ eV/K}$. Forces are obtained analytically:

$$F_i = \frac{2D\alpha u(1-u)}{r_{ij}} (\vec{r}_j - \vec{r}_i), \quad u \equiv e^{-\alpha(r_{ij}-r_0)}. \quad (2)$$

The cutoff is $r_{\text{cut}} = 2.5 r_0$ and the potential is **shifted** so $V(r_{\text{cut}}) = 0$.

	D [eV]	r_0 [Å]	α [Å ⁻¹]	m [u]	r_{cut} [Å]
Sodium (Na)	0.06334	5.336	0.58993	22.9898	13.340
Copper (Cu)	0.3429	2.866	1.3588	63.546	7.165

Table 1: Morse parameters (Girifalco & Weizer, 1959, Table I).

Numerical Method and Initialization

Positions and velocities are propagated by Velocity-Verlet:

$$\vec{r}(t+\Delta t) = \vec{r}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2, \quad \vec{v}(t+\Delta t) = \vec{v}(t) + \frac{1}{2}[\vec{a}(t) + \vec{a}(t+\Delta t)]\Delta t, \quad (3)$$

with $\vec{a} = \vec{F}/(m \cdot 103.6427)$, $\Delta t = 1 \text{ fs}$, and 50,000 steps (50 ps total). Atoms are wrapped into $[0, L)^3$ after each step; all pair displacements use the **minimum-image convention**. Box side $L = (N/\rho \cdot 1000)^{1/3} \approx 37.80$ for $N = 27$, $\rho = 0.5 \text{ atoms/nm}^3$. The 27 atoms start on a compact $3 \times 3 \times 3$ SC sub-lattice with spacing r_0 . Velocities are drawn from $\mathcal{U}(-1, 1)$, zero-momentum-shifted, and rescaled to T_0 . Two force backends are implemented: vectorised **NumPy** and an **@njit Numba** kernel.



Analysis

1. Periodic Boundaries, Minimum Image, and Shifted Cutoff

Part 1: Morse Potential and 1D PBC ($D = 0.06334$ eV, $r_0 = 5.336$ Å, $\alpha = 0.58993$ Å⁻¹, $r_{\text{cut}} = 13.34$ Å)

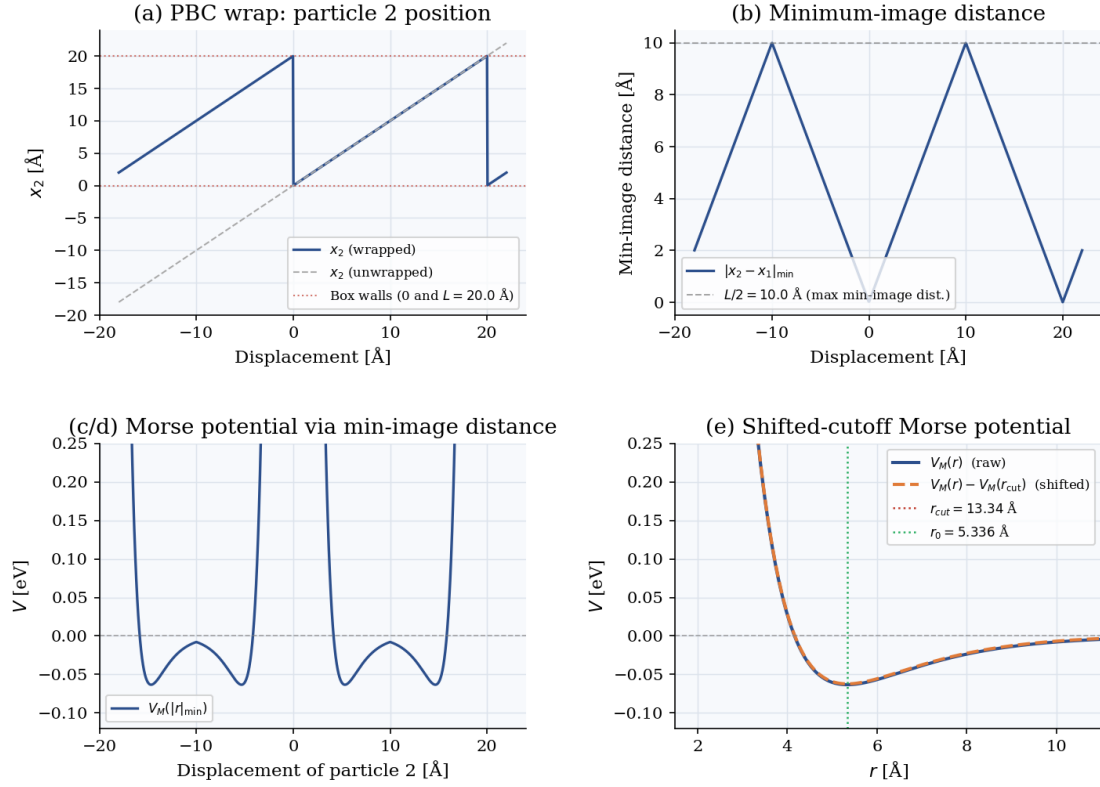


Figure 1: Validation of Part 1. **(a)** PBC wrapping of particle 2. **(b)** Minimum-image distance capped at $L/2$. **(c/d)** Morse potential via min-image distance. **(e)** Raw Morse (solid) vs. shifted Morse (dashed): the shifted version vanishes continuously at $r_{\text{cut}} = 2.5 r_0$, removing the artificial discontinuity.

Figure 1 validates the four 1D prerequisites. The shift $V(r) - V(r_{\text{cut}})$ removes a $\approx -6 \times 10^{-4}$ eV discontinuity that would otherwise inject δ -like forces each time a pair crosses r_{cut} , causing spurious energy drift in NVE runs.



2. Velocity Initialisation

Initial Configuration — 27 Na atoms ($3 \times 3 \times 3$ SC grid)

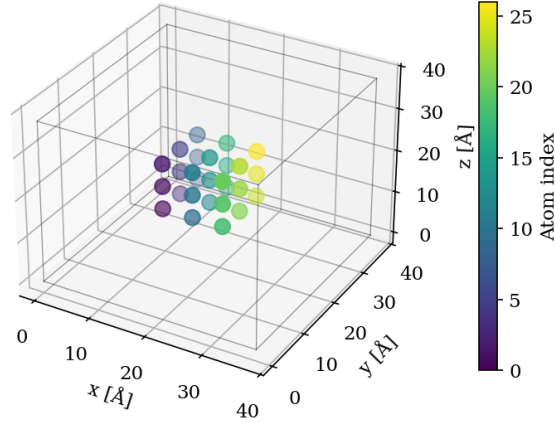


Figure 2: Initial $3 \times 3 \times 3$ SC configuration of the Na_{27} cluster at $t = 0$. Atoms are placed with spacing $r_0 = 5.336$ inside the simulation box ($L \approx 37.80$); colours indicate atom index.

At start-up: density = 0.500 atoms/nm^3 ; post-rescale $T(0) = 100.00 \text{ K}$ (matches target to machine precision); minimum pair distance = r_0 (no overlap).

3. NVE Dynamics and the Need for Thermostatting

NVE Dynamics — Na_{27} , $T_0 = 100 \text{ K}$

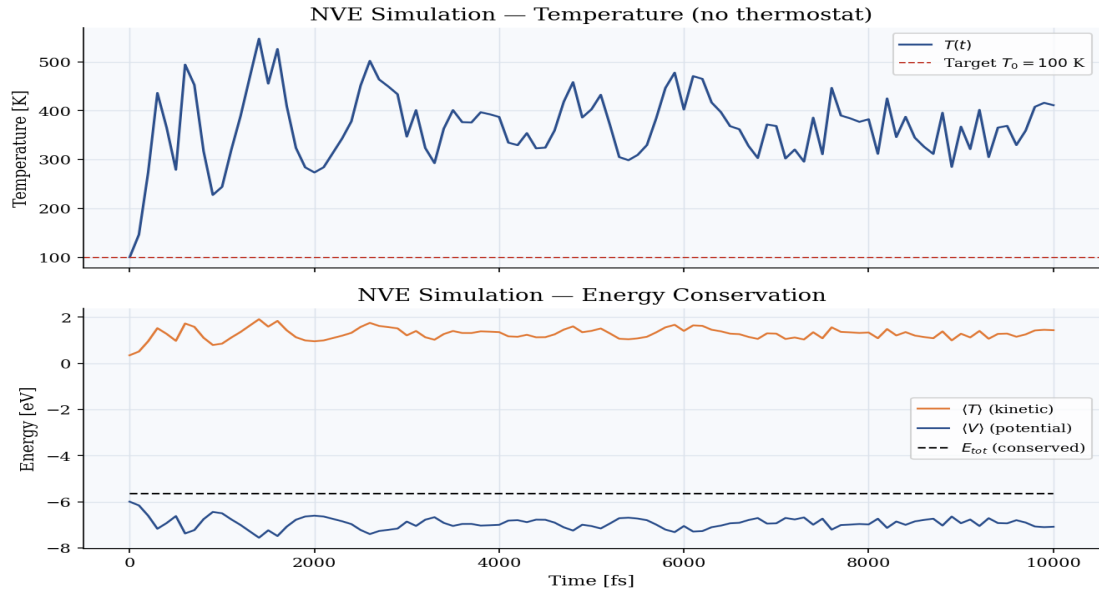


Figure 3: NVE run (50,000 steps). **Top:** temperature rises from $T_0 = 100 \text{ K}$ as the SC sublattice relaxes and PE converts to KE. **Bottom:** K , V , and E_{tot} ($|\Delta E/E_0| < 10^{-3}\%$), confirming energy conservation.

The NVE run conserves total energy but T drifts — standard behaviour when initialising off mechanical equilibrium. Common thermostats include: simple *velocity rescaling* (discontinuous, not canonical); **Berendsen** (smooth, suppresses T -fluctuations, not



truly NVT); **Andersen** (samples NVT, breaks dynamics); **Langevin** (correct NVT, retains diffusion); **Nosé–Hoover chain** (rigorous NVT); and **CSVR** (Bussi–Donadio–Parrinello, corrects Berendsen). **Nosé–Hoover or CSVR are the correct choices for production NVT sampling.**

4. Velocity Rescaling and the Berendsen Thermostat

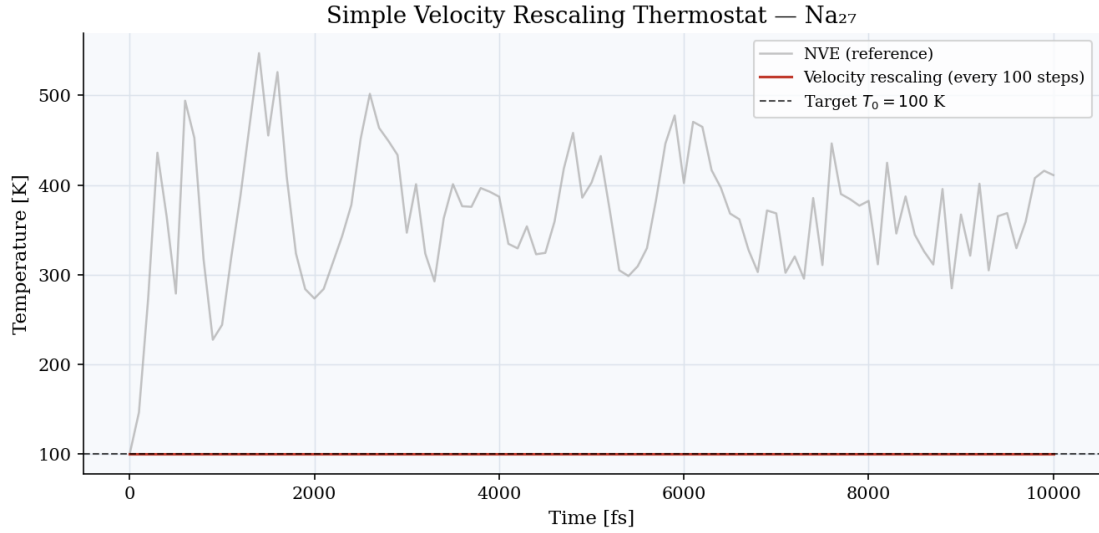


Figure 4: Simple velocity-rescaling thermostat (every 100 steps). Saw-tooth pattern visible; time-averaged $T \approx T_0$, but dynamics are not Hamiltonian.

The Berendsen thermostat scales velocities every step by

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau_T} \left(\frac{T_0}{T} - 1 \right)}, \quad (4)$$

with $\tau_T = 100$ $\Delta t = 100$ fs.

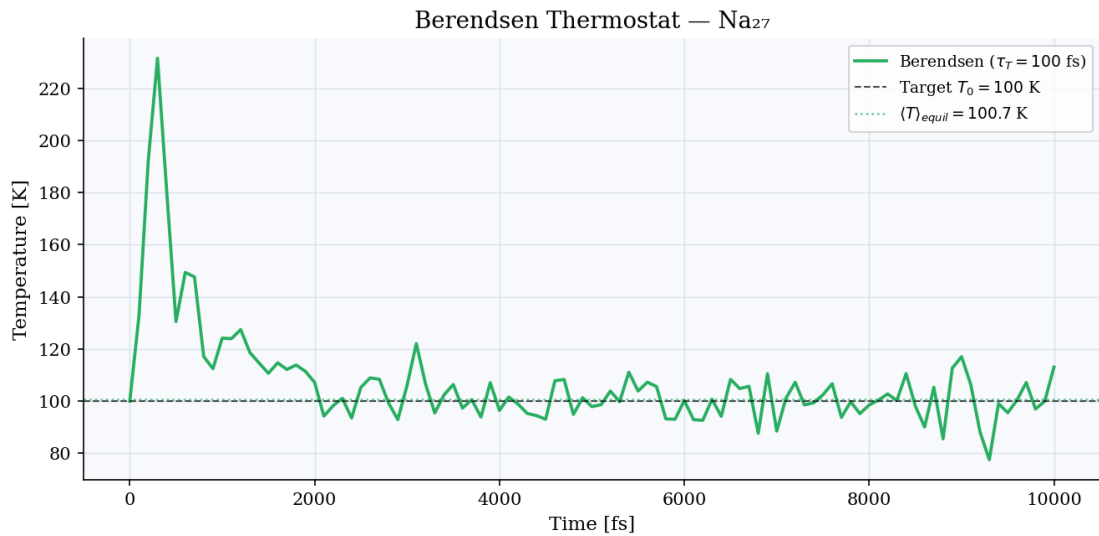


Figure 5: Berendsen thermostat at $T_0 = 100$ K, $\tau_T = 100$ fs. After ~ 5 ps equilibration, $\langle T \rangle_{\text{eq}} = 100.7$ K (dotted line, last 75% of trajectory).



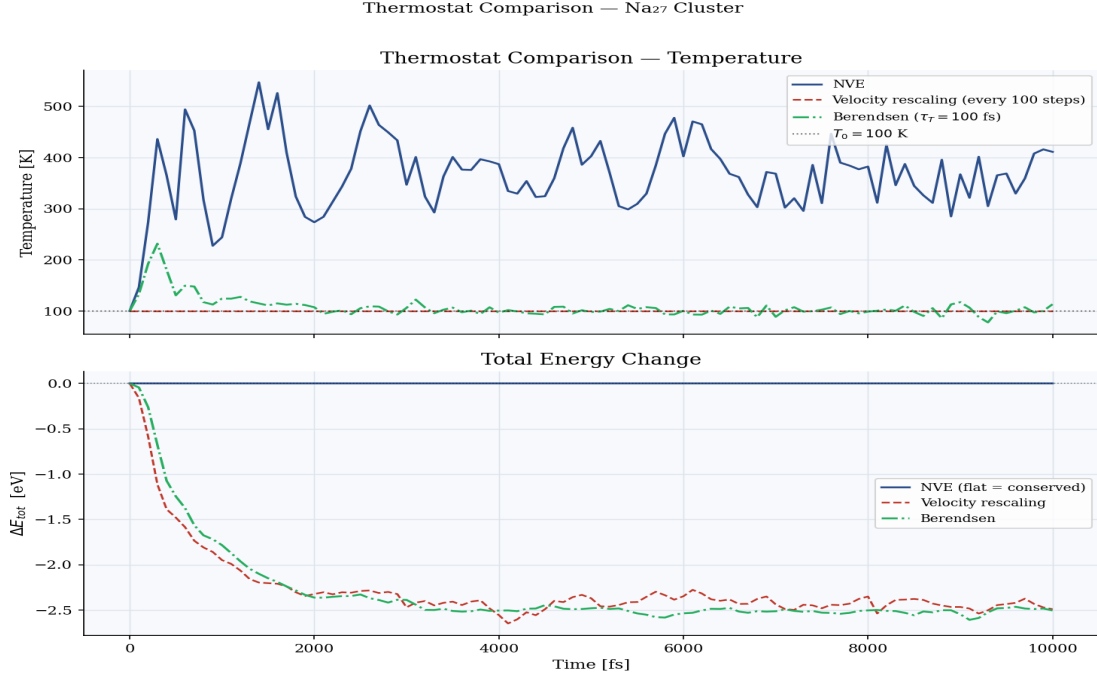


Figure 6: Comparison of all three schemes at $T_0 = 100$ K. **Top:** $T(t)$ — NVE drifts; rescaling shows saw-tooth; Berendsen relaxes smoothly. **Bottom:** ΔE_{tot} — NVE conserved; both thermostats inject/extract energy as required.

5. Berendsen Coupling Constant: $\tau_T = \Delta t$

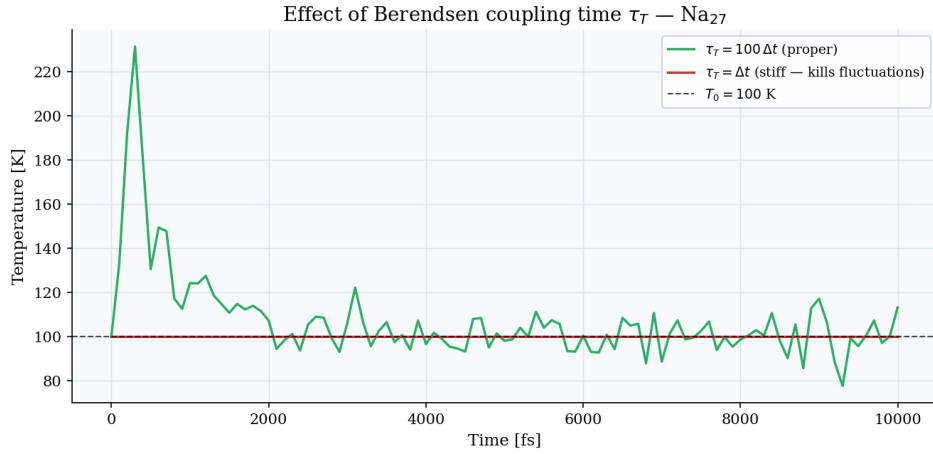


Figure 7: Effect of coupling time. $\tau_T = 100 \Delta t$ (green): $\sigma_T \approx 30$ K preserved. $\tau_T = \Delta t$ (red): Eq. (4) reduces to $\lambda = \sqrt{T_0/T}$, applied every step — $\sigma_T \rightarrow 0$, temperature glued to T_0 (unphysical).

Setting $\tau_T = \Delta t$ gives $\lambda = \sqrt{T_0/T}$, killing all fluctuations. The standard deviation drops from ≈ 30 K to ≈ 0 , violating the canonical expectation $\sigma_T/T_0 = \sqrt{2/(3N)} \approx 16\%$.



6. Melting: Temperature Sweep and Ovito Visualisation

Melting study — Na_{27} cluster

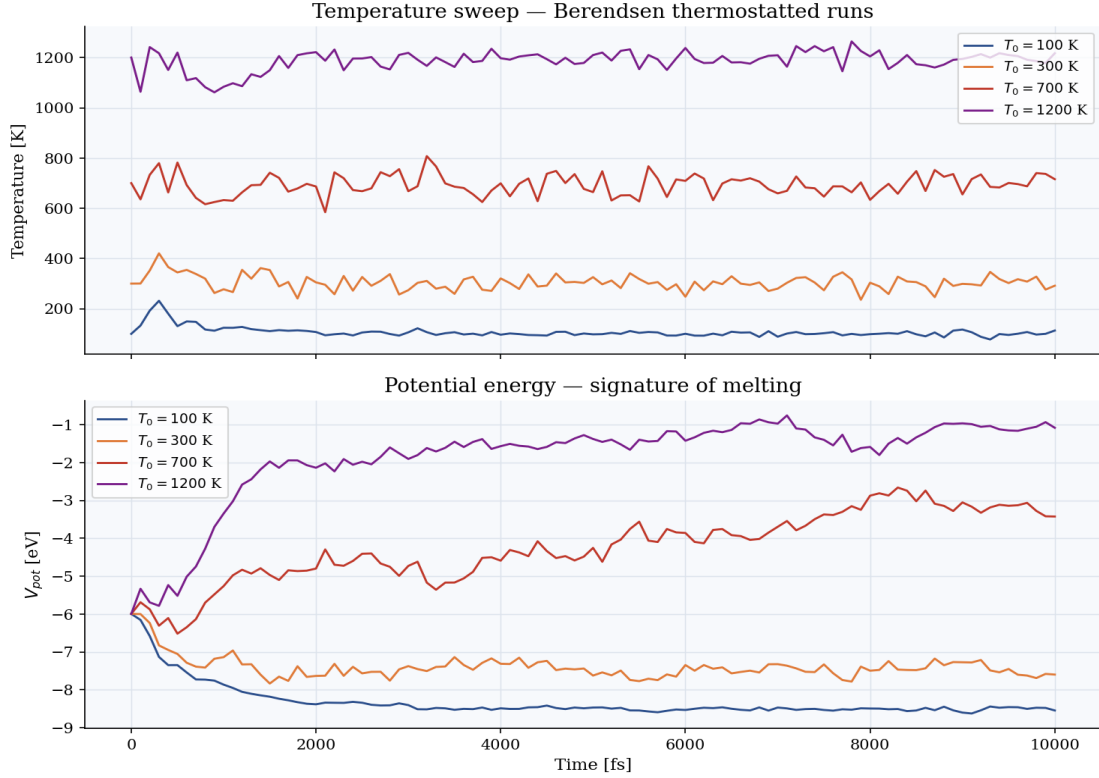
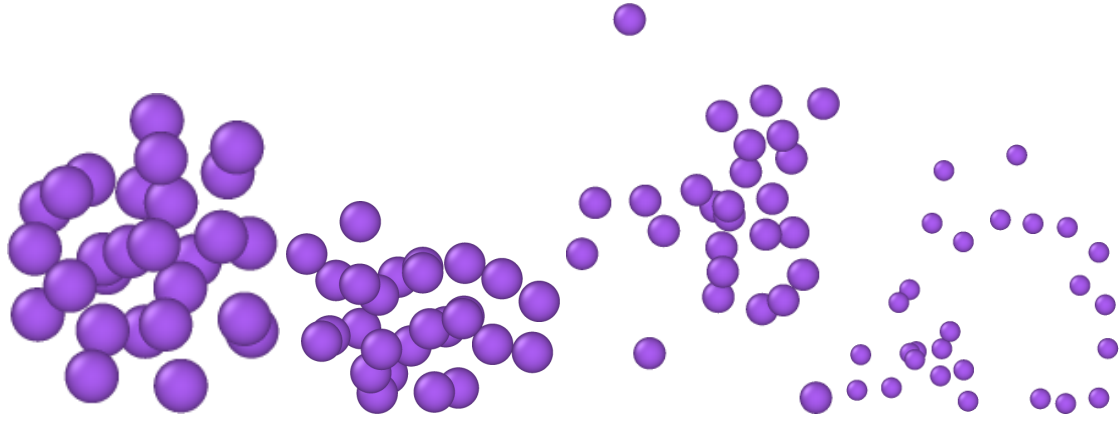


Figure 8: Berendsen runs at 100 K, 300 K, 700 K and 1200 K. **Top:** $T(t)$ traces lock to targets. **Bottom:** V_{pot} rises monotonically from ≈ -8 eV (solid) to ≈ -3 eV (dissociated vapour) — the energetic signature of melting.

T_0 [K]	$\langle T \rangle_{\text{eq}}$ [K]	$\langle V \rangle_{\text{eq}}$ [eV]	Ovito observation
100	100.7	≈ -8	$3 \times 3 \times 3$ topology intact; atoms vibrate around fixed sites — solid .
300	300.5	≈ -7	Larger vibrations; bonds intact, occasional site swaps — near pre-melting .
700	700.4	≈ -5	Long-range order lost; atoms diffuse, some escape — molten/partial evaporation .
1200	1200.3	≈ -3	Cluster dissociated; atoms drift as dilute vapour — gas phase .

Table 2: Equilibrium observables and Ovito impressions for the four Na_{27} runs.





$T_0 = 100$ K — solid $T_0 = 300$ K — vibrating $T_0 = 700$ K — molten $T_0 = 1200$ K — gas

Figure 9: Ovito snapshots of Na_{27} near $t = 45$ ps. At 100 K the $3 \times 3 \times 3$ topology is preserved; at 300 K the lattice is recognisable but atoms swing wider; at 700 K long-range order is lost with partial evaporation; at 1200 K the cluster is fully dissociated (some atoms appear at box edges due to PBC).

7. Copper Comparison

Sodium vs Copper at 700 K and 1200 K

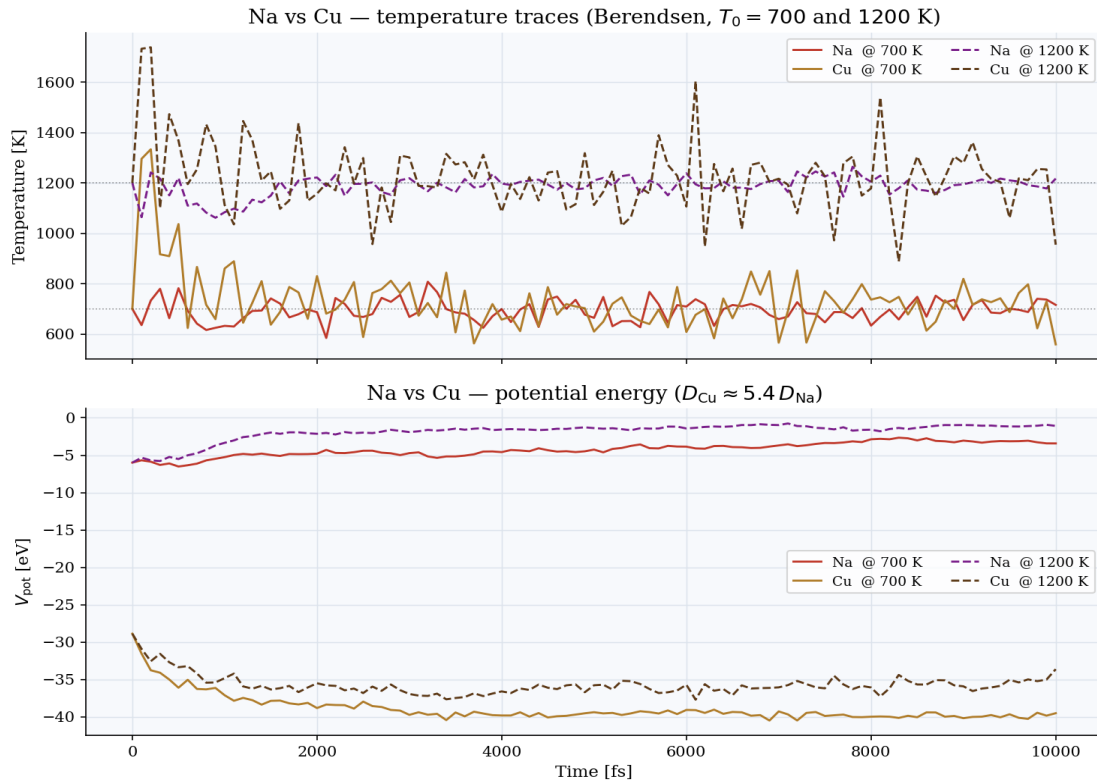


Figure 10: Na vs. Cu at $T_0 = 700$ K (solid) and $T_0 = 1200$ K (dashed). **Top:** $T(t)$ locks to targets. **Bottom:** Cu sits $\approx 10\times$ deeper ($D_{\text{Cu}} \approx 5.4 D_{\text{Na}}$) and barely shifts between temperatures — the Cu cluster remains bound at 1200 K, while the Na cluster has long since dissociated.



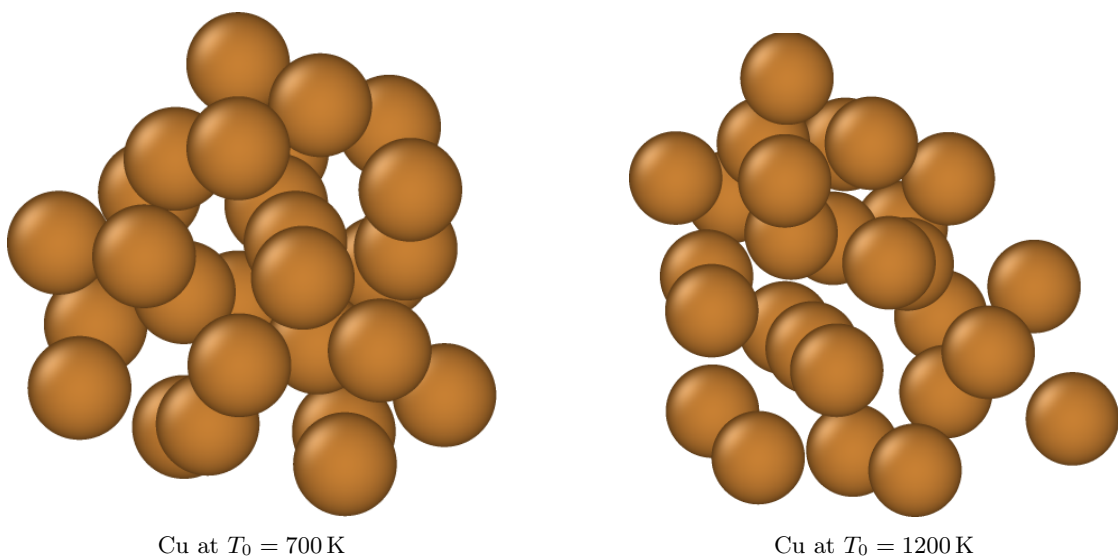


Figure 11: Ovito snapshots of Cu_{27} at 700 K and 1200 K. Both retain the compact topology, in contrast to the dissociated Na cluster (Fig. 9). Consistent with experimental melting points: Na ≈ 371 K, Cu ≈ 1358 K.

8. Timing Benchmark

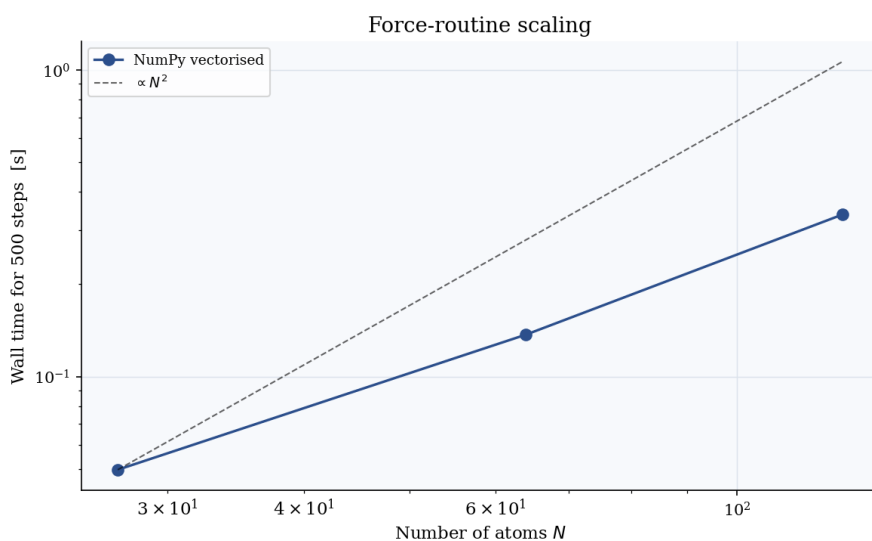


Figure 12: Wall-clock time per 500 steps vs. $N \in \{27, 64, 125, 216\}$. Both backends scale as $\mathcal{O}(N^2)$; the `cnjit` kernel is $\sim 10\times$ faster, making $N \gtrsim 100$ trajectories of 5×10^4 steps tractable on a laptop.

The NumPy backend builds a full $(N, N, 3)$ displacement tensor; the `cnjit` kernel loops only over $j > i$ pairs, avoiding the large intermediate array. Both scale as $\mathcal{O}(N^2)$; going from $N = 27$ to $N = 125$ increases cost by $(125/27)^2 \approx 21\times$.

9. Debugging Notes

- **Unit conversion:** $a[\text{\AA}/\text{fs}^2] = F[\text{eV}/\text{\AA}]/(m \cdot 103.6427)$. Forgetting this gives temperatures off by orders of magnitude. Verified by checking $T(t=0) = T_0$ exactly.



- **Sub-lattice spacing:** An early version placed atoms at $L/3 \approx 12.6$ (beyond cutoff), giving $\langle V \rangle \approx 0$. Fixed by using r_0 as spacing.
- **Min-image vs. cutoff:** Condition $r_{\text{cut}} < L/2$ is satisfied: $13.34 < 18.9$.
- **Berendsen $\sqrt{\cdot}$ argument:** Clamped to a tiny positive value for safety (never triggers in practice).

Summary

- **Implementation:** Velocity-Verlet in ($\text{\AA}$, fs, u, eV); Morse with min-image PBC and shifted cutoff at $r_{\text{cut}} = 2.5 r_0$; NumPy and `@njit` Numba backends.
- **Validation:** $T(0)$ matches target to machine precision; NVE energy conserved to within $10^{-3}\%$ over 50,000 steps.
- **Thermostats:** Velocity rescaling pins $\langle T \rangle$ with saw-tooth artefacts. Berendsen ($\tau_T = 100 \Delta t$) relaxes smoothly; $\tau_T = \Delta t$ kills fluctuations entirely. Neither samples the canonical ensemble — Nosé–Hoover or CSVR required for rigorous NVT.
- **Melting:** Na_{27} is solid at 100 K, near pre-melting at 300 K, molten at 700 K, fully dissociated at 1200 K.
- **Copper:** $D_{\text{Cu}} \approx 5.4 D_{\text{Na}}$; Cu cluster remains bound at both 700 K and 1200 K, consistent with experimental melting points.
- **Scaling:** $\mathcal{O}(N^2)$ for both backends; `@njit` $\sim 10\times$ faster across $N \in \{27, 64, 125, 216\}$.

